OR-AS Operations Research Applications and Solutions	Case Name: Office Finishing Works (4)	Sector	Construction (commercial building)	
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources	
			Schedule with costs	
		Risk Analysis	Random simulation	
			One of nine std. scenarios	
Submitted by	Pieter Van Schoors		User defined distributions	
Date	2013	Project Control	Automatic tracking	
File Name	C2013-16 Office Finishing Works (4)		Tracking based on user input	

1. Project description

Project authenticity

The execution of interior finishing works for an office building, including installation, surface treatments, and final detailing.

The project consists of activity, resource, and cost data obtained directly from the actual project execution.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	8
Planned Duration (PD)	196d
Budget At Completion (BAC)	248,203.92€
Renewable Resources	-
Consumable Resources	-

Network topology	
Serial/Parallel (SP)	33%
Activity Distribution (AD)	50%
Length of Arcs (LA)	12%
Topological Float (TF)	63%

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	19	36	2.52
CRI-rho	34	34	1.24
CRI-tau	45	48	0.38

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	42	52	0.38
SI	53	46	-0.02
SSI	33	42	0.53
CRI-r	22	31	1.50
CRI-rho	35	29	0.26
CRI-tau	44	43	0.53

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	26.46	-1.77
PV - SPI	45.12	17.87
PV - SCI	45.51	17.36
ED - 1	183.29	153.24
ED - SPI	45.12	17.87
ED - SCI	45.12	17.72
ES - 1	5.57	-4.75
ES - SPI(t)	19.20	18.38
ES - SCI(t)	18.98	18.23

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	0.89	0.12
CPI	0.24	0.00
SPI	15.12	15.12
SPI(t)	10.10	10.10
SCI	15.10	15.10
SCI(t)	10.06	10.06
0.8 CPI + 0.2 SPI	13.23	13.23
0.8 CPI + 0.2 SPI(t)	4.45	4.45

According to the MAPE values[1] the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over **5** tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	27230.20	-6295.34	28.45	1.12	0.96	1.23	0.98
std dev	29001.45	9359.84	39.02	0.22	0.05	0.38	0.03
final	49636.92	556.39	65	1.25	1.00	1.50	1.00

2.3.3.2. Time forecasting

PD	196d	Real Duration	108d	Late/early	44%

EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	261.40	7.50	142.04	-142.04
PV - SPI	264.80	10.03	145.19	-145.19
PV - SCI	249.80	46.61	131.30	-131.30
ED - 1	259.40	3.97	140.19	-140.19
ED - SPI	264.80	10.03	145.19	-145.19
ED - SCI	261.20	30.75	141.85	-141.85
ES - 1	227.80	43.85	110.93	-110.93
ES - SPI(t)	231.00	46.89	113.89	-113.89
ES - SCI(t)	233.80	54.98	116.48	-116.48

2.3.3.3. Cost forecasting

BAC	248,203.92€	Real Cost	198,567.00€	Over/under budget	20%

EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	220973.72	29001.45	12	-11
CPI	230291.82	57628.65	16	-16
SPI	225223.78	32112.70	14	-13
SPI(t)	223063.73	30030.86	13	-12
SCI	234484.66	59983.64	18	-18
SCI(t)	232224.40	57905.95	17	-17
0.8 CPI + 0.2 SPI	228460.94	50716.60	15	-15
0.8 CPI + 0.2 SPI(t)	228020.86	50226.33	15	-15